Maple Thermodynamics Academy - Phase A blueprint

Edition: 2027 Division B | Historical Phase A plan | Public revision: 2026-10-18.

Visibility: public-student. This historical plan retains the original instructional sequence. Regional Release 1.0 now contains Units 1–5, 33 chapters, 33 worksheets, and four 24-question tests. Units 6–7, WS34–WS47, and quizzes remain deferred; the earlier plans below are not claims of completed content. Detailed rules-derived passages have been excluded conservatively because a reliable comparison with the authorized manual was unavailable. Use official resources for competition decisions.

Learning destination

A student should be able to explain energy transfer with a particle model, select and rearrange an appropriate equation, track units, interpret a diagram or dataset, and justify a device prediction. Correct arithmetic without physical meaning is not full mastery. Every quantitative chapter will teach at least two fully worked numerical examples. Historical chapters use evidence and conceptual connections instead of artificial calculations.

Instructional architecture

There are 47 planned chapters paired one-to-one with WS01–WS47. Each lesson contains objectives, prerequisites, conceptual teaching, definitions, equations where meaningful, confusions, applications, recap, review questions, worksheet links, and a mastery checklist. Every worksheet has eight original questions worth 20 points: four 2-point entry questions and four 3-point application questions. The final item is a challenge within the permitted mathematics. Teachers may split a worksheet across practice and homework. No lesson needs advanced algebra, calculus, or trigonometry.

The repeated numerical solution format is Given → Find → Equation → Substitution → Arithmetic → Answer with units → Physical interpretation. Assessments sample explanations as well as calculations; open reasoning is manually graded. Suggested mastery threshold: 16/20, with correction of any misconception involving heat, Kelvin, or conservation of energy before advancing. This is a local teaching criterion, not official tournament scoring.

Progression and chapter specifications

Unit 1 builds vocabulary before formulas: quantities → boundaries → property types → equilibrium → temperature scales → energy units. Unit 2 then connects particle arrangements to phase changes and reads curves before calculating sensible/latent heat or using PV=nRT. Unit 3 distinguishes the mechanisms before conductivity and calorimetry. Unit 4 introduces a fixed sign convention before processes and engines. Unit 5 ties each scientist to ideas already encountered; historical vignettes can also be inserted earlier. Unit 6 is an explicitly optional extension for State/National preparation. Unit 7 uses the earlier physics to analyze data, predictions, and official scoring. A short device-data warm-up can begin after Unit 3; full scoring waits until students understand ratios and absolute values.

Quantitative and scientific guardrails

- Heat Q is energy crossing a boundary because of a temperature difference. Internal energy U is a system property. Do not describe heat as a substance stored in a cup.

- Use $\Delta U = Q - W$, with Q positive into the system and W positive when done BY the system. If another source uses work ON the system, explicitly translate signs.

- Use Kelvin for PV=nRT, Stefan–Boltzmann calculations, and absolute-temperature efficiency ratios. A Celsius temperature change has the same numerical size as a Kelvin change.

- Ideal-gas internal energy depends only on temperature. Isothermal therefore means $\Delta U=0$ for an ideal gas; it does not generally mean $Q=0$. Adiabatic means $Q=0$, not fixed temperature.

- Constant-pressure work $W=P\Delta V$ and rectangular P–V areas are permitted applications. No integration, logarithmic isothermal-work formula, or trigonometric geometry is required.

- Use $Q=mc\Delta T$ for a single phase with stated constant c and $Q=mL$ for a phase change under specified conditions. A pure substance at fixed pressure has a temperature plateau during an equilibrium phase transition; do not universalize that to every mixture.

- Teach heat capacity C in J/K, specific heat c in J/(kg·K), conductivity k in W/(m·K), and radiant exitance M in W/m² as distinct quantities.

- Radiation exists at all levels as a heat-transfer mechanism. Radiant exitance, detailed blackbody radiation, Stefan–Boltzmann law, and third law are State/National-only written-test topics.

- Entropy is introduced through energy spreading and the direction of spontaneous processes. Avoid treating it solely as visible mess. The third-law statement concerns a perfect crystal approaching absolute zero.

- Use stated idealizations and provided constants. Retain intermediate precision, round the final answer, and use units on numerical work. Digital exact grading is restricted to suitable objective items with explicit tolerance.

Assessment plan

Academy assessments are instructional practice. Their scope and timing are teaching choices, not a substitute for current official competition requirements.

QZ01 (10 questions, 15 min): Units 1 vocabulary and conversions; QZ02 (10, 15 min): phases and gases; QZ03 (10, 15 min): transfer and calorimetry; QZ04 (10, 20 min): laws and processes; QZ05 (10, 15 min): history and cumulative reasoning; QZ06 (8 core + 2 optional advanced, 20 min): device data and extension. All require separate keys.

T01 (14 questions, 25 min): foundations and phases. T02 (14, 25 min): transfer/calculations. T03 (14, 30 min): processes/laws/engines. T04 (15 core + 3 optional advanced, 30 min): cumulative competition preparation. T04 core will distribute three questions to each of the five required written areas and embed device contexts where appropriate. Extra advanced questions cover the sixth area. Each test key will map missed concepts back to worksheet IDs. No quiz or test is finished in Batch 1.

Device learning and official resources

The academy teaches how energy transfer, material properties, and measurement uncertainty affect warming. Students can compare clearly labeled simulated datasets, identify assumptions, and explain differences between predictions and observations. These are scientific investigations, not instructions for constructing an approved competition device.

Detailed event dimensions, equipment specifications, timing procedures, scoring weights, penalties, and tie-break rules are deliberately omitted from this public edition. Consult the current official materials and event organizers before making competition decisions. The academy's score sandbox illustrates weighted normalization only and is not official tournament scoring.

School experiments involving hot equipment require adult supervision and school safety procedures. Printed or simulated data provide a way to study the physics without operating heating equipment at home.

Official resources:

- Current event information and access to rules: https://www.soinc.org

- Rules corrections: https://www.soinc.org/rules-corrections

- Copyright and use policy: https://www.soinc.org/copyrights-media-and-use

Maple Thermodynamics Academy is an independent educational project. It is not an official Science Olympiad publication and is not endorsed by Science Olympiad, Inc. The official rules manual is not redistributed here.

Suggested multiweek sequence

Use 70-minute meetings: 5-minute retrieval, 15-minute teaching, 10-minute modeled example, 20-minute partner practice, 10-minute discussion and corrections, 5-minute exit check, 5-minute planning. Split dense material rather than rush all 47 worksheets in meetings. Assign selected follow-up practice only after instruction. Adjust dates around school breaks; no holiday deadlines are assumed.

Sessions	Focus	Assigned resources	Checkpoint
1–3	Fundamentals	WS01–06, prerequisite diagnostic	QZ01
4–6	Matter, curves, energy, ideal gases	WS07–13	QZ02; T01
7–9	Transfer and calorimetry	WS14–20	QZ03; T02
10–13	First law, processes, cycles, engines	WS21–28	QZ04; T03
14–15	Historical synthesis	WS29–33	QZ05
16–17	State/National extension, if needed	WS34–37	Advanced practice
18–21	Device models, prediction, scoring	WS38–47	QZ06
22	Timed cumulative preparation and correction	T04	Individual remediation

For a Regional-only sequence, omit sessions 16–17 and use that time for cumulative practice. Insert brief historical stories and device-data discussions earlier, but preserve the prerequisite chain. A 60-minute meeting trims partner practice to 15 minutes and discussion to 5; an 80-minute meeting adds 10 minutes of error analysis. These are teaching sessions, not an event-length simulation.

Export and privacy decisions

The master prompt requests both answer-bearing structured records and a public export without keys. Resolve this by retaining complete assessment records (answers, rubrics, numeric tolerance, explanation) only in private-instructor. Public JSON contains the corresponding question IDs and student-facing data without answers or grading metadata. Do not ship a client-side answer store under a hidden route. Client-side downloaded keys are public even if a button calls them teacher-only. An instructor-only storage folder is an organization convention, not authentication.

The root content_manifest.json describes finished resources and planned IDs. Only completed records have paths; null planned paths are not broken links. Publish only public-student. Root planning/handoff files are for the builder. The student manifest lists public resources alone. Math is UTF-8 plain text with Unicode symbols; numeric values and units also appear separately in structured records. Graph/table descriptions remain accessible without a renderer. Student PDF downloads contain no key material. The official manual is not included in either archive.

Production gates

Phase A: rules audit, full 47-row coverage matrix, scope labels and export specification.

Phase B: seven units, chapter/worksheet/key pairs, content and calculation checks for each batch.

Phase C: six quizzes, four tests, complete teacher keys, formula packet, glossary, seven review sheets, coach sessions, diagnostic, device exercises and multieweek sequence.

Phase D: validate all JSON, identifiers, question counts, paths, key correspondence, privacy, print rendering and archive contents. A batch can pass its local checks without completing the full curriculum. PROJECT_STATUS.md must keep those claims separate.